MASON DUBOEF

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Website · LinkedIn · GitHub

EDUCATION

University of Massachusets Amherst *Dec 2026 (expected)*
M.S. in Computer Science
GPA: 3.94/4.0
Relevant courses: Reinforcement Learning | AI Alignment | Machine Learning | Artificial Intelligence | Algorithms, Game Theory & Fairness | Cyber Effects | Linear Algebra

Rensselaer Polytechnic Institute (RPI) *June 2023*
B.S. in Computer Science (Concentration in AI & Data)
Relevant courses: Intro to AI (later mentored) | Economics & Computation | Intro to Network Science | Foundations of CS (Theory) | Intro to Algorithms | Data Structures | Multivariable Calculus & Matrix Algebra | Differential Equations | Operating Systems | Prog. Languages | Computability & Logic

RESEARCH INTERESTS

I am broadly interested in reinforcement learning, including work in alignment and partial observability.

RESEARCH EXPERIENCE

Distributionally Aligned Contrastive Preference Learning *Feb 2026 - Present*
Supervisor: Prof. Scott Neikum SCALAR Lab - UMass Amherst

- Learning directly from preference data while maintaining diverse dominant strategies

Algorithmic Fair Allocation for Food Rescue *Sep 2025 - Dec 2025*
Supervisor: Prof. Yair Zick FED Lab - UMass Amherst

- Produced fair and efficient allocation of food under changing dynamics and stochastic availability
- Integer linear program optimized routing of drivers from food donors to receiving agencies

Interpretable Prediction & Large-Scale Analysis of Judging in Boxing *Jan 2024 - Present*
Supervisor: Dr. Allan Svejstrup Nielsen Jabbr

- Developed “AI boxing judge” with agreement rate comparable to top-level judges
- Supervised learning mapped stats output by computer vision system onto judges’ scores
- Evaluation of top judges and analysis of their stylistic differences
- Oral presentation at *2026 MIT Sloan Sports Analytics Conference* as a competition finalist

Stable Matching in OPRA Voting Platform *Sep 2022 - Jun 2023*
Supervisor: Prof. Lirong Xia RPI

- Back-end Django development on OPRA, an online preference reporting and aggregation system
- Added support for stable matching problems, including deployment of different matching algorithms

PUBLICATION

M. duBoef, T. Romeas, M. Charbonneau, & A. S. Nielsen. *Interpretable Prediction and Large-Scale Analysis of Judging in Professional Boxing.* MIT Sloan Sports Analytics Conference Research Paper Competition (MIT SSAC), March 2026. [**Finalist – Top 7 among 197 submissions**]

PROFESSIONAL EXPERIENCE

Jabbr - Research Intern *Jan 2024 - Present*
Reference: Dr. Allan Svejstrup Nielsen (CEO)

- ML-based research on judging in professional boxing.

Mammoth Media - Data Science Intern *Aug 2021 - Jan 2022*
References: Solene Schwartz (COO) and Zachary Chow (General Manager)

- Statistical analysis of TikTok ads, informing branded content creation & media buying strategy

Meta - Data Challenge Finalist *Apr 2021 - Aug 2021*

- Four-month training program at Meta, mentored by Facebook data scientists and engineers

Luum.io - Software Engineering Intern *May 2020 - Aug 2020*
Reference: Sebastien Gouin-Davis (CEO)

- Web and android development for lighting control platform

PERSONAL PROJECTS

Multi-Objective Preference Inference from Demonstration *March 2026 - Present*

- Action-by-action Bayesian inference of demonstrator's unknown preferences over multiple objectives

NYC Subway Challenge *Oct 2025 – Dec 2025*

- Hierarchical reinforcement learning to find a route for the NYC Subway Challenge (a minimum spanning walk problem) using value iteration and between-ness clustering

Willow *Mar 2023 - Jun 2023*

- Working under Prof. Bram Van Heuveln (RPI)
- Expanded web app used to build and assign truth trees, enabling Davis-Putnam type logic problems

Have I Been Gerrymandered? *Sep 2022 – Dec 2022*

- An interactive online map indicating how gerrymandered individual congressional districts are
- Developed novel extension to efficiency gap, a measure of district fairness given electoral data

Dynamic Subway Tolling for Congestion Deterrence *Mar 2022 - Jun 2022*

- Devised optimal toll pricing to deter congestion and promote efficiency on NYC's 1 Line
- Used Nash equilibrium analysis and traffic simulation based on MTA data

Automatic Door Control *Jun 2021 - Aug 2021*

- Android development and circuit design for accessibility project, enabling remote opening of doors

SKILLS

Languages	Python, C/C++, Java, Django, SQL, TypeScript, Kotlin, React.js, Prolog, Lisp, Haskell, & MIPS Assembly
Tools	TensorFlow, PyTorch, Sklearn, Pandas, Numpy, Git, Matplotlib, Jupyter Notebook, Google Colab, Gurobi, Postman, Gephi, LaTeX, Beautiful Soup & WordPress